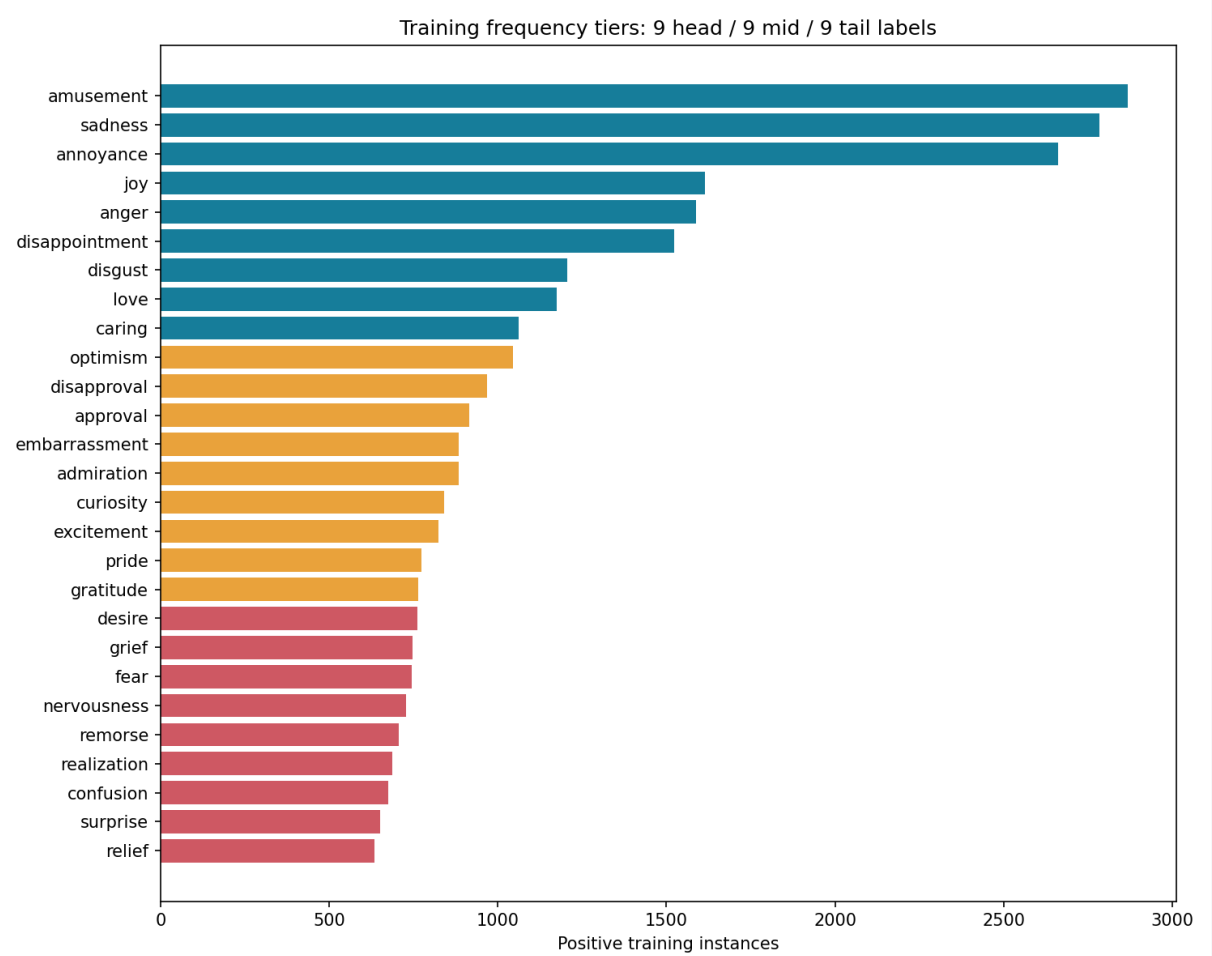
ViGoEmotions: Task and Dataset

- Multi-label emotion recognition in Vietnamese comments.
- 20,664 comments; 27 emotions plus a Neutral state.
- Effective train / validation / test: 16,530 / 2,066 / 2,067.
- Three models; compare 27 and 28 outputs.
- All model scores are validation results.

Data Profile and Leakage Prevention

- Mean 1.859 emotions; median 11 tokens / 47 characters.
- Amusement–relief frequency ratio: about 4.52×.
- Fixed split; one overlapping training sample removed.
- Fit vocabulary/IDF on train only; no model test scores.



27 vs 28 Outputs: What Changes?

- 27 outputs: Neutral-only has an all-zero emotion target.
- 28 outputs: Neutral becomes a separate learned label.
- Keep mixed annotations; no Neutral exclusion rule.
- Validation: 75 Neutral-only + 19 mixed = 94.
- Independent sigmoid outputs with binary cross-entropy.

Models and Shared Training Protocol

- TF-IDF + OvR LR: liblinear; max_iter = 1,000.
- BiGRU from scratch: hidden 128; batch 64; lr 0.001.
- PhoBERT full fine-tuning: batch 16; lr 2e-5.
- Both neural models: 20 epochs per label setting; seed 42.
- Select checkpoints by Macro-F1 on the same 27 emotions.

Results: 27 Emotions

- TF-IDF Macro-F1: 0.2388.
- BiGRU Macro-F1: 0.4220.
- PhoBERT Macro-F1: 0.5974; Micro-F1: 0.5996.
- PhoBERT Hamming Loss: 0.0550; LRAP: 0.7419.
- Higher F1/LRAP is better; lower Hamming Loss is better.

Results: 28 Labels Including Neutral

- TF-IDF Macro-F1: 0.2310.
- BiGRU Macro-F1: 0.4086.
- PhoBERT Macro-F1: 0.5783; Micro-F1: 0.5829.
- PhoBERT Hamming Loss: 0.0575; LRAP: 0.7077.
- These metrics evaluate all 28 labels.

Does Neutral Help the Shared 27 Emotions?

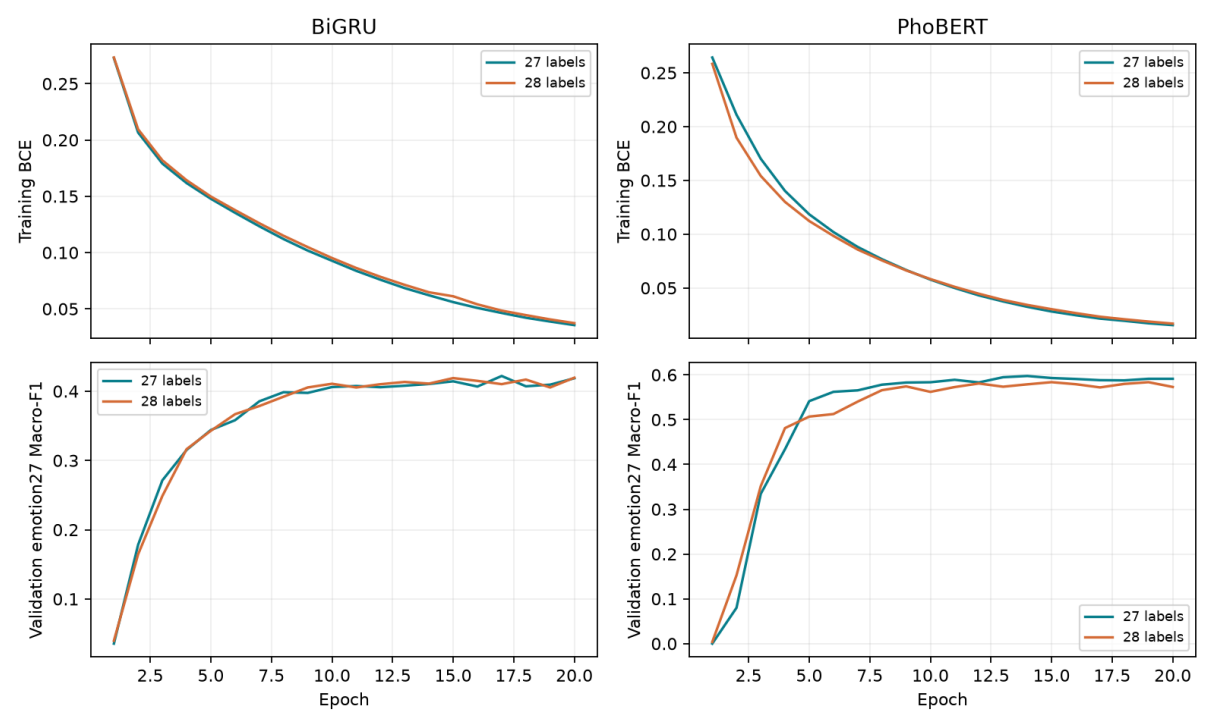
- PhoBERT with 27 outputs: 0.5974.
- PhoBERT with 28 outputs, scored on 27: 0.5836.
- Difference: -1.38 percentage points.
- No observed improvement on the shared emotions.
- One seed; no causal or statistical conclusion.

Neutral Detection Remains Difficult

- PhoBERT Neutral F1: 0.4369.
- Precision: 0.4018; recall: 0.4787.
- 45 TP / 67 FP / 49 FN; support: 94.
- The 27-output model has no Neutral classifier.
- An empty prediction is not automatically Neutral.

Learning Curves and Best Checkpoints

- All four neural runs completed epochs 1–20.
- Best BiGRU epochs, 27/28: 17 / 20.
- Best PhoBERT epochs, 27/28: 14 / 15.
- Select the best checkpoint, not necessarily the last.
- Training loss alone cannot confirm overfitting.



Q1–Q2: Per-Label Performance and Frequency

- PhoBERT best label: gratitude, F1 0.8370.
- Lowest label: disapproval, F1 0.3623.
- Training frequency vs F1: Spearman rho 0.018.
- p-value 0.928; descriptive evidence only.
- Complete per-label results are in the report.

Q3–Q4: Label Pairs and Text Styles

- Top training pair: disappointment–sadness (904 cases).
- Both correct: 63/122 validation cases; joint recall 51.64%.
- Largest PhoBERT–BiGRU style gap: teencode.
- Macro-F1 gap: +22.13 percentage points.
- Style groups are heuristic and may overlap.

Q5: Does the Default Threshold Matter?

- PhoBERT 27 at 0.5: Macro-F1 0.5974.
- Best validation grid point: 0.35, F1 0.5978.
- Same checkpoint; exploratory validation analysis.
- Very small F1 gain; Hamming Loss increases.
- Main results remain at threshold 0.5.

Error Analysis: New Neutral Predictions

- Six cases: three Neutral false positives and three misses.
- Potential issues: implied emotion, short text, HTML, profanity.
- IDs and abstract observations; no raw comments.
- Preliminary AI-assisted review; team confirmation pending.

Final Research Question and Next Steps

- Is one threshold suitable for every emotion?
- Freeze the best neural checkpoint and its weights.
- Compare 0.5, global tuned and per-label thresholds.
- Tune on validation; lock choices; evaluate test once.
- Review at least 20 false negatives and 20 false positives.

Conclusions and Evidence

- PhoBERT leads within both label settings.
- Neutral does not improve shared-emotion F1 in this run.
- Evidence: Kaggle 08/09, saved predictions and manifests.
- Limits: one seed, validation selection, no test scores.
- Sources: course brief, ViGoEmotions, PhoBERT and libraries.